

NETWORKED CONTROL

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1) Consider the continuous-time system

$$\dot{x} = 4u$$

$$y = x$$

- A. Assuming that the system is controlled using a **continuous-time control law** of the type $u(t) = Kx(t)$, compute the range of values of K that make the closed-loop system asymptotically stable.
- B. Assuming that the system is controlled using a **digital control law** (i.e., with a sample-and-hold actuator) of the type $u_k = Kx_k$ and with sampling time $h = 0.1$ s, compute the range of values of K that make the closed-loop system asymptotically stable.
- C. Assume that the system is controlled using the **networked control system** sketched below.

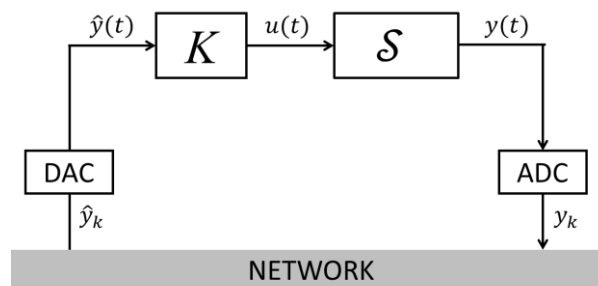


Figure 1: Networked control system

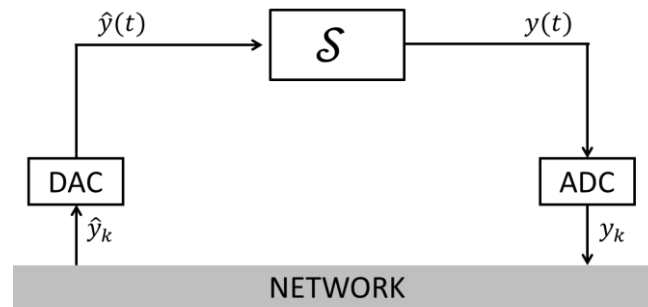
Assume that the sampling time is constant and equal to $h = 0.1$ s. Define the range of values of K that make the closed-loop system asymptotically stable in case the **network delay is constant** in the following three cases:

- a. $\tau = 0.025$ s;
 - b. $\tau = 0.05$ s;
 - c. $\tau = 0.075$ s.
- D. Assume again that the system is controlled using the networked control system sketched above, with constant sampling time and gain $K = -1$. Compute the **maximum allowable transmission interval** in the following two cases:
- a. $\tau = 0$ s
 - b. $\tau = 0.05$ s

2) Define in details the **delays in the network channel**, commenting the equations that allow to define the propagation and frame time.

What are the typical **bit rates**, **message sizes**, and **frame times** of the most widespread control networks (e.g., Ethernet, Devicenet)?

3) Consider a **networked control system** of the type described by the following figure.



where the continuous-time dynamics of system S is described by

$$\mathcal{S}: \begin{cases} \dot{x}(t) &= Ax(t) + B\hat{y}(t) \\ y(t) &= Cx(t) \end{cases}$$

Assume that both the **sampling time** h_k and the **network delay** τ_k are time-varying. Derive, possibly defining an "augmented" state vector, the discrete-time model describing the **networked control system dynamics**.